

The Morphophonology of Reduplicated Presents in Vedic and Indo-European*

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1. Introduction and aims

The verbal system of Vedic Sanskrit employs a diverse array of reduplicated formations that are indispensable for the reconstruction of reduplicated verbal categories in Proto-Indo-European; see Kulikov 2005:431–5 and citations therein for a linguistically current résumé of all Vedic verbal formations with reduplication. Among these formations, the reduplicated present (Class III in the traditional nomenclature of Sanskrit present classes; cf. Whitney 1889:227–78) shows the greatest diversity of vocalism in its reduplicating syllable.¹ Specifically, reduplicated presents exhibit a variation between *i*- and *a*-vocalism in the reduplicant that lacks an immediate phonological or morphological explanation. For most roots that form reduplicated presents in the Vedic period, the reduplicant vocalism is stable by lexeme; for instance, $\sqrt{dā}$ ‘give’ and $\sqrt{bhas}$ ‘chew’ build 3.sg.pres.act.ind. forms *dādāti* and *bābhasti*, whereas $\sqrt{bhr}$ ‘bear’ and $\sqrt{vac}$ ‘speak’ build 3.sg.pres.act.ind. forms *bíbharti* and *vívakti*, respectively. Nevertheless, a few roots show an alternation between *i*- and *a*-vocalism in the reduplicant, e.g., 3.sg.pres.act.ind. *síṣakti* but 3.pl. *sáṣcati* to $\sqrt{sac}$ ‘follow’, or 3.sg.pres.act.ind. *jígāti* but (fossilized) pres.part.act. *jágat-* to $\sqrt{gā}$ ‘go’.

Although reduplicated presents are non-productive already from the earliest phase of the Vedic language, for the reduplicated present stem to be a synchronically non-derivable, lexicalized fossil is unhappy, because the burden of accounting for all the peculiarities of its inflection then falls on the prehistory of the formation. Precisely in this vein, some scholarship (e.g., LIV:17, Tichy 2006:113–4, and Jasanoff 2003:128–32) projects the distinction between Vedic

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1 Henceforth, I will use the terms “reduplicant” and “base” to refer to what one might otherwise designate as the “reduplicating syllable” and the “root.” I ignore any theoretical implications that the former set of terms may have.

a-reduplicating and *i*-reduplicating presents back to distinct PIE types with **e*- and **i*-reduplication. Such a reconstruction simplifies the morphological and phonological problems of inflection for the Vedic reduplicated present, but introduces a problem of semantic distinction between the two types; furthermore, the reconstruction of two PIE types does not account for the *u*-reduplication found in Indo-Iranian in a direct way.

The approach and objective of this paper are to attempt a systematic morpho-phonological account of the reduplicated present in Vedic that can capture the variations in reduplicant vocalism at the synchronic level. I will further discuss the implications of the reduplicative process at work in Vedic for

- theories of reduplication situated within the larger framework of generative phonology

and

- the reconstruction of the reduplicated present for Proto-Indo-European.

This paper is not, strictly, an historical account of every detail in the inflection and attestation of Vedic reduplicated presents. Rather, this work primarily seeks to identify a theoretical model of reduplication that can generate the Vedic data synchronically, regardless of precisely how well a given stem is attested in the language (cf. notes to table 1); secondarily, this paper suggests how one might best reconstruct the reduplicated present to account for certain forms that are not susceptible to the synchronic analysis.

2. Inflectional and phonological classification of the Vedic reduplicated present

This section introduces the sources of data on reduplicated presents in Vedic employed for this study, the criteria for classification of verbal forms as genuine reduplicated presents, and a classificatory scheme of the reduplicated presents.

2.1. Data

I have drawn data concerning the inflection of the Vedic reduplicated present from the following sources:

- the *Ṛgveda* (RV), via the concordance of Lubotsky (1998);
- the Śaunaka Recension of the *Atharvaveda* (AV), via the index verborum of Whitney (1881);

- lemmata of Whitney 1885, for forms from Vedic texts other than the above-mentioned.

Relatively few relevant forms are attested outside of the RV or AV, and indeed many traces of reduplicated present inflection for a given root are entirely limited to the RV. Whitney (1885:212–3) lists forty-nine roots as attesting to the stem of an athematic reduplicated present. However, Whitney's classifications include some forms that are properly to be understood as pluperfects or modal forms to a perfect stem. I apply the following criteria to determine whether a given root has a reduplicated present:

1. If an unmistakable perfect stem is attested:
 - (a) forms with primary endings, active or middle, in indicative mood, clearly indicate a reduplicated present stem, even if ultimately derivative of a perfect;
 - (b) if no forms with primary endings, or only modal forms are attested, those forms probably belong to the perfect stem;
 - (c) if the perfect stem to the root shows *a*-reduplication, a stem with *i*-reduplication may reflect a reduplicated present stem.
2. If no unmistakable perfect stem is attested, then an imperfect or modal form may constitute evidence of a reduplicated present.

Based on these criteria, one should exclude $\sqrt{dh\bar{i}}$ 'think', $\sqrt{p\bar{i}}$ 'swell', $\sqrt{vyac}$ 'surround', and probably $\sqrt{man_2}$ 'wait'² on the basis of criterion 1(b). At the same time, however, criterion 1(a) then demands that one consider *bibhēti* (to $\sqrt{bh\bar{i}}$ 'fear') and *mamatsi* (to $\sqrt{mad}$ 'enjoy') as reduplicated presents, despite the fact that such forms do ultimately derive from perfects; cf. Cardona 1992 and Insler 1972.³ Table 1 presents those roots that attest to the synchronic formation of athematic reduplicated presents.

2 Provided that the pluperfects and modal forms with the sense 'wait' ultimately derive from $\sqrt{man_1}$ 'think'; see the discussion in Kümmel 2000:364–6. If one insists that these forms belong to a separate root, then one may consider these forms as attesting to a reduplicated present based on criterion 2.

3 Indeed, that these derived reduplicated presents seem to conform to the morphophonological specifications for reduplicated presents given below suggests that the reduplicated present was not an absolutely closed and lexicalized class in the early Vedic period.

Table 1. Athematic reduplicated presents

Skt. root	Gloss ^a	PIE root ^b	Past pass. part.	Class III example
√ <i>r</i>	'move'	* <i>h₃er-</i>	X	<i>īyarti</i>
√ <i>gā</i>	'go'	* <i>g^weh₂-</i>	X	<i>jīgāti</i>
√ <i>ghr</i>	'drip'	* <i>g^wher-</i>	<i>ghrtá-</i>	<i>jígharti</i>
√ <i>ci</i>	'observe'	* <i>k^weǵ-</i>	<i>citá-</i>	2.sg.pres.act.ind. <i>cikéṣi</i> ^{AV}
√ <i>juṣ</i>	'enjoy'	* <i>ǵeus-</i>	<i>júṣta-, juṣtá-</i>	3.pl.pres.act.subj. <i>jujuṣan</i> R
√ <i>tij</i>	'be sharp'	* <i>(s)teǵ-</i>	<i>tiktá</i> ^V	2.sg.pres.act.impv. <i>titigdhi</i> ^{MS} H
√ <i>dā</i>	'give'	* <i>deh₃-</i>	<i>dattá-, -dāta-,</i> <i>-tta-</i>	<i>dádāti</i>
√ <i>diś</i>	'point'	* <i>deǵk-</i>	<i>diṣtá-</i>	3.sg.impf.mid.inj. <i>didiṣta</i>
√ <i>dhā</i>	'put'	* <i>d^heh₁-</i>	<i>hitá-, -dhita-</i>	<i>dádhāti</i>
√ <i>dhr</i>	'hold'	* <i>d^her-</i>	<i>dhrtá-</i>	2.pl.pres.act.impv. <i>didhrtá</i>
√ <i>nij</i>	'wash'	* <i>neǵ^w-</i>	<i>niktá-</i>	2.pl.pres.act.impv. <i>ninikta</i> H
√ <i>nī</i>	'lead'	* <i>neǵH-</i>	<i>nītá-</i>	2.du.pres.act.ind. <i>ninīthás</i>
√ <i>pr</i>	'carry, keep'	* <i>per-</i>	X	<i>píparti</i>
√ <i>prc</i>	'mix'	* <i>perk-</i>	<i>prktá-</i>	2.sg.pres.act.impv. <i>piprgdhi</i> R
√ <i>bhas</i>	'chew'	* <i>b^hes-</i>	<i>bhasita</i> ^C	<i>bábhasti</i>
√ <i>bhī</i>	'fear'	* <i>b^heǵh₂-</i>	<i>bhūtá-</i>	<i>bíbhéti</i> ^V
√ <i>bhr</i>	'bear'	* <i>b^her-</i>	<i>bhrtá-</i>	<i>bíbharti</i>
√ <i>mad</i>	'enjoy'	* <i>med-</i>	<i>mattá</i> ^{AV}	2.sg.pres.act.ind. <i>mamatsi</i> R? or H

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Notes: X = unattested; R = rare (i.e., the stem is known from three or fewer forms), H = hapax legomenon. All forms are RV and 3.sg.pres.act.ind. unless otherwise marked: AV = Atharva-veda, MS = Maitrāyaṇī Saṃhitā, V = Vedas, C = Classical Sanskrit. Accents are marked only when attested.

a Glosses generally follow Whitney (1885) and Kümmel (2000).

b Forms of PIE roots follow LIV.

Table 1—Continued

Skt. root	Gloss	PIE root	Past pass. part.	Class III example
√mā ₁	'measure'	*meh ₁ -	mita- ^{AV}	3.sg.pres.act.impv. mimātu
√mā ₂	'low'	X	X	mīmāti
√mī	'(ex)change'	*mei ₁ H	mīta- ^V	3.du.pres.act.ind. mīmītas R
√yas	'(be) heat(ed), boil'	*ies-	yastá- ^V	3.sg.pres.act.impv. yayastu H
√yu	'separate'	*iey-	yutá-	yuyoti
√rā	'give'	*reh ₁ -	rātá-	2.sg.pres.act.impv. rirīhi
√vac	'speak'	*uek ^w -	uktá-	vīvakti
√vaś	'wish, want'	*uek-	X	vivaṣṭi H, 2.sg.pres. act.ind. vavákṣi H
√viṣ	'handle'	*ueis-	viṣta-	viveṣṭi
√śā	'sharpen'	*keh ₃ -	śitá-	śiśāti
√sac	'follow'	*sek ^w -	X	siṣakti, 3.pl.pres.act. ind. sáscati
√sas	'sleep'	*ses-	X	sasásti ^{TS} R
√su	'press'	*sey-	sutá-	3.pl.pres.act.ind. suṣvati H
√sr	'let loose'	*sel-	srtá- ^B	sísarti
√has	'laugh'	*g ^w hes-	hasita- ^E	pres.part.act. jákṣat- H
√hā	'leave behind, depart'	*ĝ ^h eh ₁ -	hāna- ^{AB} , hīná-, jahitá-	jáhāti, 3.sg.pres.mid. ind. jīhīte
√hu	'pour, offer'	*ĝ ^h ey-	hutá-	juhóti
√hū	'call'	*ĝ ^h eyH-	hūtá-	1.pl.pres.act.ind. juhūmási R
√hrī	'be ashamed'	*ĝ ^h rej ₁ H-	hrīta- ^B	jihreti ^B

Notes: TS = Taittirīya Saṃhitā, AB = Aitareya Brāhmaṇa, B = Brāhmaṇas, E = Epic Sanskrit.

Some forms given here require further comment:

- *jujuṣan* (RV 7.61.6d) to √juṣ could easily be a perfect subjunctive, though all other RVic perfect subjunctives to this root show full grade of the base. Also notable is 2.pl.pres/perf.act.impv. *jujuṣtana*. The forms' interpretation as reduplicated presents thus mostly relies on the etymological connection with Hittite *ku-ku-uš-zi* 'tastes' and Young Avestan

3.sg.pres.mid.ind. ($\bar{a}$ -)zūzušte; cf. Watkins 2003:391. *LIV*:166–7, on the other hand, does not believe that any Indo-Iranian forms continue an old reduplicated present.

- *titigdhi* to $\sqrt{tij}$ is a nonce form, though with *LIV*:593 “im Kontext fungiert *titigdhi* eindeutig als zu *téjas*- n. ‘Schärfe, Glut, Energie’ gehöriger Präsensstamm.” *titigdhi* is a synchronically valid Class III present.
- The imperatives *didhrtam* and *didhrtá* most probably belong to the better attested reduplicated aorist stem *dīdhar*-. *didhrtam* (RV 5.86.6ef) should scan with long $\bar{i}$ in order to fit the lines’ cadences; *didhrtá* (RV 1.139.8g) may scan with short *i*.
- Note that, pace Whitney (1885:100), $\sqrt{p\bar{r}}$ ‘fill’ (< **pleh₁*-) probably does not have a reduplicated present. Having read all the occurrences of the stem *pīpar*-/ *pīpr*- in the RV and AV Śaunaka, I find that only AV Śaunaka 1.35.4b (*pīparmi*) lends itself well to an interpretation ‘fill’. Furthermore, as Whitney (1905:520–1, 636) already mentions, the AV Paippalāda sometimes reads *bibharti* where AVŚ has *pīparti*.
- Both *vivaṣṭi* (RV 7.16.11b) and *vavákṣi* (RV 8.45.6b) are problematic. *EWA*:527 identifies both forms as “Augenblicksbildungen”; Joachim (1978:151–2) likewise suggests that both forms are nonces; *LIV*:672–3, misreading Joachim 1978, identifies only *vivaṣṭi* as an Augenblicksbildung, and tentatively supports a link between *vavákṣi* and Hittite *we-wakki*; Lubotsky (1998:1246) marks *vavákṣi* as a nonce, but not *vivaṣṭi*. A 3.sg.pres.act.impv. *vivaṣtu* is found in the Sāmaveda, but in a mantra which is a borrowing of RV 7.16.11. According to the synchronic principles of formation developed in §4 below, *vavákṣi* has aberrant reduplicant vocalism, but could preserve an archaic accentuation; on the other hand, *vavákṣi* could be built directly to a perfect stem, with perfect accentuation, like *mamáttu* to $\sqrt{mad}$, but the perfect of $\sqrt{vaś}$ is rare (3× RV), and in all cases shows a long reduplication vowel, thus *vāvaś*–; see further Kümmel 2000:477–80. In my assessment, the stem *vivaś*– is more likely the synchronically “grammatical” Class III formation, but that Vedic inherited any reduplicated present to $\sqrt{vaś}$ is very doubtful.

Table 2 displays those reduplicated presents that also show thematization, but exhibit phonological patterns that indicate the synchronic productivity of a reduplicated present stem:

Table 2. Thematized reduplicated presents

Skt. root	Gloss	PIE root	Past pass. part.	Class III example
√ <i>ghrā</i>	‘sniff, smell’	* <i>g^{wh}reh₁-</i>	<i>ghrātá</i> ^B	3.pl.pres.act.ind. <i>jighranti</i> ^{MS} R
√ <i>nas</i>	‘return’	* <i>nes-</i>	X	3.sg.pres.mid.ind. <i>nĩmsate</i>
√ <i>pad</i>	‘go, step’	* <i>ped-</i>	<i>panná</i> ^{AV}	pres.mid.part. <i>pĩbdamāna-</i> H
√ <i>myakṣ</i>	‘affix’	* <i>meĩk̑-(s)-</i>	X	<i>mimikṣati</i> R
√ <i>sthā</i>	‘stand’	* <i>steh₂-</i>	<i>sthitá-</i>	<i>tĩṣṭhati</i>
√ <i>han</i>	‘strike’	* <i>g^{wh}en-</i>	<i>hatá-</i>	3.sg.pres.mid.ind. <i>jighnate</i>

Notes: see notes to table 1.

Whether the reduplicated present stem to √*ghrā* is thematic or athematic is not entirely clear. Whitney 1885:43 reads: “jigrati AV.MS (3p).” However, Whitney (1905:694) acknowledges that the form in the AV (12.4.5) “can be ... either singular or plural.” Furthermore, Amano (2009:556n2426) follows the emendation of von Schroeder (1881) in reading the MS manuscript form (*prati*)*jighrati* as (*prati*)*jighranti*. Kāṭhaka Saṃhitā forms (e.g., 3.sg.pres.act.opt. (*ava*)*jighret*, 2.sg.pres.act.impv. (*ava*)*jighra*) also suggest thematic inflection.

Table 3 shows further forms that must historically continue reduplicated presents, but in which sound changes either eliminated a clear morphological division separating the reduplicant and the base or obscured an obvious phonological relation between reduplicant and base.

Table 3. Historical reduplicated presents

Skt. root	Gloss	PIE root	Past pass. part.	Class III example
√ <i>aj</i> > √ <i>ij</i>	‘drive’	* <i>h₂eǵ-</i>	<i>-ajita</i> ^{SB} H	3.sg.pres.mid.ind. <i>ĩjate</i> R
√ <i>r</i> > √ <i>ir</i>	‘move’	* <i>h₃er-</i>	<i>ĩrṇá</i> ^B	3.sg.pres.mid.ind. <i>ĩrte</i> (cf. <i>iyarti</i> in table 1 above, to which <i>ĩrte</i> remains the corresponding mid- dle in early Vedic)

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Notes: SB = Śatapatha Brāhmaṇa. See notes to table 1.

Table 3—Continued

Skt. root	Gloss	PIE root	Past pass. part.	Class III example
√īḍ	'praise'	*h ₂ eṛsd-	īḍitá-	3.sg.pres.mid.ind. íṭṭe
√ih	'be eager'	*Heṛġh-	ihita ^E	2.sg.pres.mid.ind. ihase ^{VS} H
√pā	'drink'	*peh ₃ -	pītá-	píbati
√sad > √sīd	'sit'	*sed-	sanná- ^{AV}	sīdati

Notes: VS = Vājasaneyi Saṃhitā. See notes to table 1.

Vedic speakers had probably reanalyzed these forms as either root (Class II) or thematic (Class I) presents to entirely different roots. One need not, therefore, bring to account this last group from the synchronic point of view, though these forms may contribute data to the Indo-European picture.

2.2. Phonological patterns between root and reduplicant

When we examine the data for relationships between the vocalism of the reduplicant and vocalism or other structure of the root, the following patterns emerge:

- Roots that take a zero grade with *u*-vocalism invariably reduplicate with *u*-vocalism (e.g., *juhóti*).
- Roots that take a zero grade with *i*-vocalism invariably reduplicate with *i*-vocalism (e.g., *cikéṣi*).
- Roots that take a zero grade with *r*-vocalism invariably reduplicate with *i*-vocalism (e.g., *bíbharti*).
- Thematic reduplicated presents invariably take zero grade of the root in the base but reduplicate with *i*-vocalism (e.g., *tíṣṭhati*).
- Roots of other structures (i.e., *CaC* or *Cā*) show either *i*- or *a*-vocalism in the reduplicant, but some further subregularities are observable:
 1. *CaC*-roots in final coronal (i.e., √*bhas*, √*mad*, √*yas*, and √*sas*) always show *a*-vocalism in the reduplicant.
 2. √*vac*, √*vaś*, and √*sac* typically show *i*-vocalism in the reduplicant.⁴ Significant exceptions are the 3.pl.pres.act.ind. *sáścati* and corre-

4 See the discussion on *vivaṣṭi* and *vaváṣi* following table 1 above.

spending pres.part.act. *sáścat-* to $\sqrt{\text{sac}}$;⁵ I will take up these forms in §5.

3. *Cā*-roots (i.e., $\sqrt{\text{gā}}$, $\sqrt{\text{dā}}$, $\sqrt{\text{dhā}}$, $\sqrt{\text{mā}_1}$, $\sqrt{\text{mā}_2}$, $\sqrt{\text{rā}}$, $\sqrt{\text{śā}}$, and $\sqrt{\text{hā}}$) exhibit the greatest diversity of reduplicant vocalism: $\sqrt{\text{dā}}$ and $\sqrt{\text{dhā}}$ invariably reduplicate with *a*-vocalism; $\sqrt{\text{rā}}$ shows *a*-vocalism of the reduplicant except in the 2.sg.pres.impv.act. *rirīhi*; $\sqrt{\text{hā}}$ divides its reduplicant vocalism into *a* for the active paradigm but *i* for the middle paradigm; the remaining four roots always show *i*-vocalism of the reduplicant.

In the main, the vocalism of the reduplicant is predictable based on either a morphological criterion (if thematic, the reduplicant takes *i*-vocalism) or a phonological criterion (root structure). To that extent, these patterns constitute “Islands of Reliability” in the sense of Albright (2002). Namely, where a general morphological pattern does not predominate, phonological subregularities may come into play in order to determine the morphological expression of a category. Indeed, the vocalism of the reduplicant very closely corresponds to the vocalism of the zero-grade allomorph of a given root as it appears in the past passive participle (see tables 1–3) or other morphological category that regularly takes zero grade of the root. For instance, even in the diverse *Cā*-roots, $\sqrt{\text{śā}}$ builds a PPP *śita-* and a reduplicated present *śiśāti*, while $\sqrt{\text{dā}}$ attests a PPP *-dāta-* or *-tta-* in compounds (instead of perhaps expected **dītā-* < **dh₃-tō-*; cf. Greek *δοτός*) and a reduplicated present *dādāti*. Regardless of the precise historical origin of the reduplicated present, then, such “Islands of Reliability” suggest that, at some point in the prehistory of Vedic (perhaps Indo-Iranian), subcategories of reduplication relying on common phonological patterns established themselves. These patterns alone, however, do not constitute an account of how a Vedic reduplicated present is built in the process of morphological and phonological derivation: unless one can provide both historical and theoretical bases for these patterns, their collection here is little better than the listing of aberrant reduplicated forms found in Whitney 1889:665–6 or Macdonnell 1916:144–5.

3. Reduplication cross-linguistically

The data presented in §2 pose a difficult problem in that many approaches to reduplication, whether in the context of generative phonology or historical

5 Note that the participle is homonymous except in accent with the more common participle to $\sqrt{\text{sac}}$ ‘dry up’ (*a*-)*saścat-* (14× RV).

linguistics, presuppose that the defining relationship between base and reduplicant is of a phonological nature. See, for instance, Wilbur 1973, Marantz 1982, or even the more recent models of McCarthy and Prince 1995 and Raimy 2000. Although reduplication could be defined purely in terms of a phonological copying mechanism, such a definition would immediately exclude the Vedic reduplicated present, which does not build reduplicants of absolutely reliable phonological form. Furthermore, most reduplicative processes known cross-linguistically are morphophonological processes: the morphological marking found on a category gives phonological expression to some syntactic or semantic value, but the phonological expression of the morpheme itself depends upon independent phonological operations. A purely phonological definition of reduplication, then, would not capture an essential aspect of many reduplicated categories across languages generally. Hence, Inkelas and Zoll (2005:2) carefully distinguish between two distinct linguistic processes, which nevertheless bear some surface similarity to one another:

1. Phonological Copying
2. Morpho-Semantic Feature Reduplication [Morphological Reduplication]

In the succeeding subsections, I will define and expand upon these two categories.

3.1. *Phonological copying*

Phonological Copying entails that the phonological features of some segment are reproduced in another position in a word in order to satisfy structural demands of a purely phonological nature, e.g., that no syllable lack an onset consonant or that no syllable lack a coda consonant. Inkelas and Zoll (2005:2–3, 20–1) cite cases of this phenomenon in Hausa, Yoruba, and Spokane (Interior Salish); in examples (1)–(2) I reproduce the Spokane case, in which an affix *-e-* meaning REPETITION (REP) is added to the verbal root (data are taken from Black 1996:210ff.; cf. also Bates and Carlson 1997:105–7). If the verbal root is of the form *CC* at the stage in the derivation when *-e-* enters, the *-e-* is simply infixes; see (1a)–(b), where *šl'*- and *lč'*- become *š-e-l'*- and *l-e-č'*-. If instead the verbal root is of the form *CVC* at the stage in the derivation when *-e-* enters, the *-e-* is prefixed, and the first consonant of the root is copied to provide an onset to the syllable, thus resulting in a form *C₁VC₁VC*; see (2a)–(b), where *šəl'*- and *nič'*- become *š-e-šil'*- and *n-e-n'ič'*-.

- (1) Repetitive *-e-* infixes into an initial consonant cluster
- a. /-e-, šl'-n'-t-ən'/ → š'-e-l'n'tén'
 REP, chop-CONTROL-TRANSITIVE-1SGTRANSITIVESUBJECT → 'I cut it up repeatedly'
 - b. /-e-, lč'-n'-t-ən'/ → l'-e-č'n'tén'
 REP, tie-CONTROL-TRANSITIVE-1SGTRANSITIVESUBJECT → 'I tied it over and over'
- (2) Phonological copying provides onset for *-e-*
- a. /-e-, šəl'/ → š'-e-šil'
 REP, chop → 'I cut up repeatedly'
 - b. /-e-, nič'-n'-t-əx^w/ → n'-e-n'ičn'tx^w
 REP, cut-CONTROL-TRANSITIVE-2SGTRANSITIVESUBJECT → 'you kept on cutting it'

The seeming reduplication in the examples under (2) is merely a phonological repair to a wholly phonological problem: Spokane does not allow syllables without onsets, but rather than inserting some default consonant, the features of the nearest consonant are copied into that onset position.

3.2. *Morpho-semantic feature reduplication*

Work on reduplicative constructions that encode a syntactic or semantic value has typically identified two major variations in phonological expression:

1. Fixed Segmentism: some phonological segment of the reduplicant remains constant throughout the paradigm of that reduplicated category, regardless of the base's phonological constituency.
2. Phonological Correspondence: every segment of the reduplicant has a correspondent in the base, though the segments of the reduplicant may undergo changes through identifiable phonological processes.

Inkelas and Zoll (2005) further consider constructions with semantic identity, but no necessary phonological relationship whatsoever, i.e., pure Semantic Doubling.

3.2.1. *Fixed segmentism.* Examples of fixed segment reduplication are readily available in the form of the reduplicated presents and the perfect of Ancient Greek, which are cognate with the Vedic reduplicated present and perfect. In the reduplicated presents, one finds *ɪ*-vocalism in the reduplicant throughout the

entire paradigm, regardless of the vocalism found in the root; likewise, the Greek perfect always displays ε-vocalism in the reduplicant:

(3) Greek reduplicated present

<u>Root</u>	<u>1.sg.act.</u>	<u>1.pl.act.</u>
√θη 'put'	τίθημι	τίθεμεν
√δω 'give'	δίδωμι	δίδομεν
√μεν 'remain'	μῖννω	μῖννομεν

(4) Greek perfect

<u>Root</u>	<u>1.sg.act.</u>	<u>1.pl.act.</u>
√λειπ 'leave'	λέλοιπα	λελοίπαμεν
√πειθ 'convince'	πέποιθα	πεποίθαμεν
√φευγ 'flee'	πέφευγα	πεφεύγαμεν

Reduplication with fixed segmentism, however, is not restricted to the vocalism of the reduplicant; Tübatulabal (Uto-Aztec, California), for instance, exhibits fixed segmentism in its reduplicant consonantism:

(5) Tübatulabal (after Alderete et al. 1997:24; data from Vogelin 1958)

<u>Base form</u>	<u>Reduplicated form</u>	<u>Gloss</u>
pi:fɪn	ʔi:-bi:fɪn	'he is snoring'
le:win	ʔe:-le:win	'to pack it'
piʃika	ʔi-piʃika	'to slip'
ʔa:baʔiw	ʔa:-ʔa:baʔiw	'it is showing'

Here, the reduplicant faithfully copies the vocalism (including length) of the base, but the onset consonant of the reduplicant universally appears as a glottal stop, regardless of what the initial consonant of the base is.

3.2.2. *Phonological correspondence.* Instantiations of phonologically correspondent reduplication abound cross-linguistically, at least in comparison to fixed segment reduplication.⁶ Under the definition given above, phonologically correspondent reduplication may encompass both "full" (i.e., every segment of the base also has surface expression in the reduplication) and "partial" (i.e., the redup-

6 Alderete et al. 1997 remains the only treatment in the linguistic literature, to my knowledge, specifically to develop an account of fixed segmentism. I report anecdotally that cases of fixed segment reduplication, despite the familiarity of the Greek examples, appear rather infrequently in linguistic literature. Moreover, queries to the Graz Reduplication Database (= Hurch 2005) yielded approximately three times as many constructions without fixed segmentism as with fixed segmentism.

plicant is somehow phonologically reduced) reduplication. Example (6) shows true "full" reduplication; example (7) demonstrates that the Vedic intensive follows a pattern of "partial" reduplication:

(6) Indonesian noun pluralization (data from Cohn 1989:185)

hák-hák	'rights'
búku-búku	'books'
kərá-kərá	'monkeys'
kəcíl-kəcíl	'small-DISTRIBUTIVE'
waníta-waníta	'women'
minúman-minúman	'beverages'

Here, two phonologically identical forms (i.e., each segment in the reduplicant has an obvious co-indexation to a segment in the base, $h_1\acute{a}_2k_3$ - $h_1\acute{a}_2k_3$ 'rights') are co-affixed to produce plural semantics.

(7) Vedic intensive reduplication (data from Schaefer 1994, s.v. given roots)

Root	3.sg.	3.pl.
√ <i>diś</i> 'point'	mid. <i>dēdiṣṭe</i>	mid. <i>dēdiṣate</i>
√ <i>nū</i> 'bellow'	act. <i>nónavīti</i>	1.pl.act. <i>nonumas</i>
√ <i>vṛt</i> 'turn'	act. <i>vársvarti</i>	act. <i>vársvrtati</i>

In this case, the reduplicant (leftmost syllable) always shows the vocalism that corresponds to the full grade of that root, while the root itself undergoes ablaut, depending upon person/number and diathesis. Strictly speaking, where the root stands in the zero grade (e.g., *dēdiṣate* or *nonumas*), not every segment of the reduplicant has a correspondent in the base: in / $d_1\acute{a}_2i_3$ - $d_1i_2\acute{s}_3$ -, the *a* in the reduplicant lacks a corresponding segment in the base, while the *s* in the base lacks a corresponding segment in the reduplicant. The morphophonological derivation proceeds as follows:

(8) Derivation of 3.sg.pres.mid.intens. *dēdiś*-⁷

1. Root: / $d\acute{a}i\acute{s}$ -/
2. Semantics: intensive
3. Morphology: reduplicate for intensive → /RED- $d\acute{a}i\acute{s}$ -/
4. Phonology: fill the reduplicant following restrictions (not more than one coda consonant) → / $d\acute{a}i$ - $d\acute{a}i\acute{s}$ -/
5. Semantics: middle diathesis

7 This derivation intends no theoretical claims whatsoever; it is merely illustrative.

6. Morphology: ablaut to zero grade of the base, add desinence →
/dáj-dəjś-təj/

3.2.3. *Semantic doubling*. Inkelas and Zoll (2005:6–8) further classify certain constructions in which no phonological relationship whatsoever obtains between the constituents as reduplication:

- (9) Khmer synonym compounds (after Inkelas and Zoll 2005:8; data from Orun and Haiman:485–501)

<u>Compound</u>	<u>Constituent semantics</u>	<u>Compound semantics</u>
cah-tum	'old+mature'	'village elder'
kee-mərdək	'heritage+heritage'	'legacy'
caməj-ʔahaa(r)	'food+food'	'food'
ʔaar-kambəj	'secret+secret'	'secret'
cbah-prakat	'exact+exact'	'exact'
peel-weeliə	'time+time'	'time'

Similar is a case from Vanatuan Sye. In Vanatuan Sye, every verbal root has two phonologically distinct, but semantically identical, stem formations that cannot be phonologically derived from one another. Historically, these stems derive from a single verbal root:

- (10) Vanatuan Sye verbal allomorphs (data from Crowley 1998:83–5; see further Inkelas and Zoll 2005:52–4)

<u>Stem 1</u>	<u>Stem 2</u>	<u>Gloss</u>
evsor	amsor	'wake up'
evtit	avtit	'meet'
ocep	agkep	'fly'
ochi	aghi	'see it'
omol	amol	'fall'
oruc	anduc	'bathe'
ovoli	ampoli	'turn it'
vag	ampag	'eat'

Which stem is used depends on the morphological category being expressed, e.g., the simple future takes Stem 2, whereas imperatives take Stem 1. Most interesting is a reduplicative construction that consists in the co-affixation of the two stems:

- (11) cw-amol₂-omol₁
3PLFUT-fall₂-fall₁
'we will fall all over'

4. Modeling the Vedic reduplicated present

This section will attempt to formulate a synchronic analysis of the Vedic reduplicated present within two distinct approaches to reduplication: Base-Reduplicant Correspondence Theory (McCarthy and Prince 1995) and Morphological Doubling Theory (Inkelas and Zoll 2005). Given the divergent theoretical claims about the nature of reduplication that these two models make, the compatibility or incompatibility of the Vedic evidence may serve to provide evidence in support of or against one model or the other.

4.1. Base-Reduplicant Correspondence Theory and Morphological Doubling Theory: Definitions

4.1.1. *Base-Reduplicant Correspondence Theory.* Under Base Reduplicant Correspondence Theory (BRCT), reduplication is a prosodic phonological process in which an abstract RED(uplicant) morpheme receives its phonological specification and ultimate surface form through the interaction and ranking of four constraint relationships:

1. General I(nput)-O(utput) markedness constraints on acceptable surface form.
2. I(nput)-R(eduplicant) faithfulness: the output reduplicant should be phonologically like the input base.
3. I(nput)-B(ase) faithfulness: the output base should be phonologically like the input base.
4. B(ase)-R(eduplicant) identity: the output reduplicant should phonologically resemble the output base.

Figure 1 represents the interactions of these various relations:

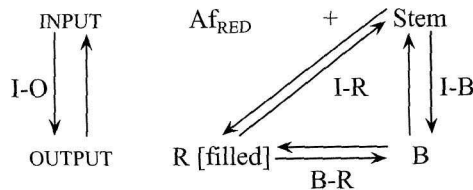


Fig. 1. The Full Model of Reduplicative Identity (after McCarthy and Prince 1995:25)

This model of reduplication depends on the theoretical claims and machinery of parallel Optimality Theory (OT).⁸

4.1.2. Morphological Doubling Theory. Morphological Doubling Theory (MDT) defines reduplication in terms of morphology and semantics: the constituents of a reduplicative construction are semantically identical, and the targets of such a construction are morphological elements (such as a word, root, stem, or affix) rather than phonological elements (cf. Inkelas and Zoll 2005:66). Thus, true reduplication, as opposed to phonological copying (cf. §3.1 above), consists in the co-affixation of two semantically identical morphemes. The fact that semantically identical morphemes are often phonologically similar, if not identical, results in the phonological similarities seen between base and reduplicant in many reduplicative constructions cross-linguistically. The exact phonological shape that a reduplicative construction adopts is the result of interactions between three cophonologies, i.e., distinct sets of phonological rules or constraints that apply to the morphological constituents independently and then at the level of affixation (cf. Inkelas and Zoll 2005:19):

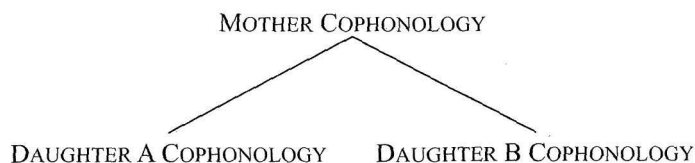


Fig. 2. Interaction of cophonologies in reduplication

I will employ Optimality Theory to represent the operation of the various cophonologies in MDT, though MDT in itself is compatible with whatever phonological framework one chooses to adopt.

4.2. Modeling under BRCT

Given that the data presented under §2 contain some patterns of reduplication in which the reduplicant and base lack an obvious relationship of identity, BRCT faces a significant challenge. The following examples will illustrate that one may formulate constraint rankings that can capture either the reduplication pattern of *Cā-* and *CaC*-roots (such as $\sqrt{śā}$ or $\sqrt{vac}$) that regularly show *i*-reduplication, or the reduplication of roots that show *a*-reduplication (e.g., $\sqrt{bhas}$ or $\sqrt{dā}$), but not

⁸ See Prince and Smolensky 1993 for an introduction to Optimality Theory.

both. Either set of constraints can correctly predict the outcome for roots with zero grades in *i*, *u*, or *r*.⁹

Since the primary concern of the following exposition lies solely in the quality of vocalism found in the reduplicant, whatever constraints necessary to produce the correct consonantism or prosodic shape will be omitted in order to simplify the exposition. I therefore take for granted the following conditions on the reduplicant for Sanskrit reduplicated presents:

- The reduplicant admits only of palatal stops (e.g., *cikéṣi*).
- All underlying aspirates surface as unaspirated [Grassmann's Law] (e.g., *bábhasti*).
- Where a root has more than one root-initial consonant, only the less sonorous of those consonants will appear in the reduplicant (e.g., *ut̥ṣṭhati*).

Limitations of space here do not permit independent justification of the constraints in use.

4.2.1. *Constraint ranking 1: MAX-IB, IDENT-IB* $\gg$ $*V_{[-high]}$ $\gg$ *IDENT-BR*. In example (12), the reduplication of $\sqrt{mā}$ 'bellow' from the full grade morphologically required for the 3.sg.pres.act.ind. surfaces as *mimāti*, in winning candidate a:

(12)

/RED-mā-ti/		MAX-IB	IDENT-IB	*R	$*\sigma_{\mu} > 1$	$*V_{[-high]}$	IDENT-BR	MAX-BR	DEP-V
a.	mi.mā.ti				*	*	**		
b.	ma.mā.ti				*	**!			
c.	mā.mā.ti				**!	*			
d.	mi.ma.ti		*!			*	**		
e.	mi.mi.ti		*!*				**		

for the following reasons:

- The IDENT-BR violations in (12a) result from a change in quality and length of vocalism in the reduplicant; however, were the reduplicant vowel to maintain the same quality and/or length, violations of higher-ranking $*\sigma_{\mu} > 1$ or $*V_{[-high]}$ would ensue; cf. candidates b and c.

9 A stipulative constraint $*R$ seems to be necessary to forbid an *r* in a reduplicant throughout, though the same restriction was probably in operation for PIE itself; see further §5.

- $*\sigma_\mu > 1$ and $*V_{[-high]}$ do not force any changes in the surface expression of the base, because MAX-IB and IDENT-IB, demanding faithfulness to the input, outrank those markedness constraints.

In example (13), the ⟨e⟩ (= [əi]) vocalism of the base is impermissible in the reduplicant according to the given constraint ranking, because it would violate both $*\sigma_\mu > 1$ and $*V_{[-high]}$; cf. candidate b.

(13)

/RED-ce-si/	MAX-IB	IDENT-IB	*R	$*\sigma_\mu > 1$	$*V_{[-high]}$	IDENT-BR	MAX-BR	DEP-V
a.  ci.ke.ṣi				*	*	*	*	
b. ce.ke.ṣi				***	**			
c. ca.ke.ṣi				*	***		*	
d. ci.ki.ṣi	*!							

However, to resolve $*\sigma_\mu > 1$ through the elimination of [i] still leaves a non-high vowel in the reduplicant, as in candidate (13c). Winning candidate (13a), on the other hand, violates low-ranking MAX-BR in deleting the [ə] and violates IDENT-BR in changing the underlying non-syllabic /i/ to syllabic [i], but thus avoids any violation of either $*\sigma_\mu > 1$ or $*V_{[-high]}$.

In order to satisfy both $*\sigma_\mu > 1$ or $*V_{[-high]}$, as well as high-ranked *R, winning candidate (14a) shows deletion of all acceptable vocalic material originally present in the base, into which an anaptyctic vowel [i] is then inserted.¹⁰

However, as examples (15) and (16) illustrate, this set of constraints predicts that roots that show *a*-vocalism in the reduplicant would instead have *i*-vocalism. Candidates (15a) and (16a) both pick out forms as winning candidates that do not exist, rather than the expected candidates of (15b) and (16b); in particular, contrast example (16) with example (12).¹¹ Therefore, this constraint ranking cannot be viable.

10 The general spread of [i] as an intermorphemic linking vowel, on analogy to original *set*-roots, could constitute a reasonable basis for assuming [i] as the standard anaptyctic vowel of Sanskrit.

11 Kulikov (2005:438n16) points out that a stem *bibhas-* is found in some manuscripts, but universally is emended to *babhas-*.

(14)

/RED-par-ti/		MAX-IB	IDENT-IB	*R	$*\sigma_\mu > 1$	$*V_{[-high]}$	IDENT-BR	MAX-BR	DEP-V
a.	पि.पर.ति				*	*		**	*
b.	पर.पर.ति				**!	**			
c.	पा.पर.ति				*	**!		*	
d.	प्र.पर.ति			*!	*	**	*	*	

(15)

/RED-bhas-ti/		MAX-IB	IDENT-IB	*R	$*\sigma_\mu > 1$	$*V_{[-high]}$	IDENT-BR	MAX-BR	DEP-V
a.	बि.भस.ति				*	*	**	*	
b.	बा.भस.ति				*	*!	**	*	
c.	बास.भस.ति				**!	*			
d.	बिस.भस.ति				**!	*	*		

(16)

/RED-dā-ti/		MAX-IB	IDENT-IB	*R	$*\sigma_\mu > 1$	$*V_{[-high]}$	IDENT-BR	MAX-BR	DEP-V
a.	दि.दा.ति				*	*	**		
b.	दा.दा.ति				*	**!	*		
c.	दादा.ति				**!	*			
d.	दि.दा.ति		*!			*	**		
e.	दि.दि.ति		*!*				**		

4.2.2. *Constraint ranking 2*: $*\sigma_\mu > 1$, $MAX-BR_{[+high]} \gg IDENT-BR$, $MAX-BR$. This constraint ranking similarly employs $*\sigma_\mu > 1$ to drive the moraic value of the reduplicant, but the removal of $*V_{[-high]}$ ensures that forms such as *bābhasti* and *dādāti* can surface properly. $MAX-BR_{[+high]}$ gives preference to high vowels in the reduplicant, thereby securing forms such as *cikēṣi* over **cakeṣi*.

Examples (17) and (18) are parallel to examples (15) and (16), demonstrating that expected *bābhasti* and *dādāti* can win using the new set of constraints:

(17)

/RED-bhas-ti/		MAX-IB	IDENT-IB	*R	$*\sigma_{\mu} > 1$	MAX-BR _[+high]	IDENT-BR	MAX-BR	DEP-V
a.	ba.bhas.ti				*			*	
b.	bi.bhas.ti				*		*!	*	
c.	bas.bhas.ti				***!				

(18)

/RED-dā-ti/		MAX-IB	IDENT-IB	*R	$*\sigma_{\mu} > 1$	MAX-BR _[+high]	IDENT-BR	MAX-BR	DEP-V
a.	da.dā.ti				*		*		
b.	dī.dā.ti				*		***!		
c.	dā.dā.ti				***!				

In (19), the faithfulness constraint MAX-BR_[+high], which establishes a bias for the preservation of [+high] segments existing in the base over [−high] segments, allows *cikéši* to surface:

(19)

/RED-ce-si/		MAX-IB	IDENT-IB	*R	$*\sigma_{\mu} > 1$	MAX-BR _[+high]	IDENT-BR	MAX-BR	DEP-V
a.	ci.ke.ši				*			*	
b.	ca.ke.ši				*	*!		*	
c.	ci.ki.ši	*!							

However, the ultimate consequence of this set of constraints is that other roots that show a type of *a*-vocalism in the base are predicted to show *a*-vocalism in the reduplicant as well, contrary to the actual data; compare (20) with (12) above:

(20)

/RED-mā-ti/		MAX-IB	IDENT-IB	*R	* $\sigma_\mu > 1$	MAX-BR _[+high]	IDENT-BR	MAX-BR	DEP-V
a.	मि.मा.ति				*		**!		
b.	मा.मा.ति				*		*		
c.	मा.मा.ति				**!				

Again, the given constraint ranking is unable to provide a successful account of all variation found in the Vedic data. Therefore, one must conclude that the theoretical machinery that BRCT offers is inadequate to explain the morphophonological patterns found in the Vedic reduplicated present.

4.3. Modeling under MDT

At least for the athematic reduplicated presents represented in table 1, MDT can provide a coherent account: the formation of a reduplicated present consists in the co-affixation of the zero-grade allomorph to a given root, as the reduplicant, to another form of the same root, taking whichever ablaut grade the inflectional paradigm specifies. The Vedic reduplicated present is akin to the reduplicative constructions of Vanatuan Sye seen in example (11) in §3.2.3—two distinct allomorphs (in some cases, synchronically non-derivable allomorphs, as in zero-grade forms with *i*-vocalism to *Cā*-roots) are bound to one another in order to fulfill the need for two semantically identical elements, similar to the Khmer synonym constructions in example (9). The derivations in examples (21)–(24) illustrate the composition of a Vedic reduplicated present: cophonologies at the appropriate daughter nodes shape the respective allomorphs for which the morphosyntax calls, and then further cycle through the mother cophonology to produce a final output.

As seen in the preceding section, the constraint $*\sigma_\mu > 1$, here present in the Daughter A cophonology, drives the prosodic shape of the reduplicant. Likewise, the constraints ranked above DEP-V result in a reduplicant of the form *Ci-* to roots with zero grades in *r*. Throughout, I take the forms of the past passive participle shown in table 1 as indicative of the morphological form of the zero grade to a given root.

- (21) Morphosyntactic operation: construct the 3.sg.pres.act.impv. for 'measure'
($\sqrt{mā}$)

		MAX-IO	
/mi-mā-/			
a.	mi-mā-		
b.	mi-m-		*!
c.	my-ā-		*!

/zero grade allomorph to 'measure'/'		*σ _H > 1	*R _i	DEP-V	MAX-IO
a.	mi-				
b.	m-				*!
c.	i-	*!			*

/full grade allomorph to 'measure'/'		MAX-IO
a.	mā-	
b.	m-	*!
c.	a-	*!

Output: *mi-mā-*
Suffix: *-tu*

→ *mimātu*

The constraint set needed to generate the correct output is very simple: Daughter B (the base) and the Mother (the co-affixed stem) need only be strictly faithful to the inputs; the reduplicant is largely faithful to the input as well, but its restricted prosodic shape induces some change. The burden for correct outputs instead falls almost entirely on the morphology, which selects the proper input form from commands received from syntax or semantics.

Given that the zero grade allomorph to $\sqrt{yas}$ is probably *yas-* as the past passive participle *yastá-* would suggest, then the *a*-vocalism found in the reduplicant of *yayastu* (22) is simply a preservation of the underlying vocalism. Given that the only free-standing form of the past passive participle to $\sqrt{dā}$ 'give' attested from the *R̥gveda* forward is *dattá-*, or *-dāta-* or *-tta-* as the second member of compounds, I submit that the zero grade allomorph that served as an input for $\sqrt{dā}$ was not $\sqrt{dā}$, as *mi-* is to $\sqrt{mā}$ (21), but rather *dad-* or *dā-*, thus making the vocalism of the reduplicant straightforward to obtain.¹²

12 Some special pleading, though, is necessary to explain *dādhāti* to $\sqrt{dhā}$, which attests only a PPP *hitá-*, or *-dhita-* as the second member of compounds and in sandhi. I see two possible factors driving the formation of *dādhāti*: (i) thoroughgoing identity of inflection to $\sqrt{dā}$, and (ii) *hitá-* as a suppletive form for the PPP taken from a lower register of speech. If one or both

- (22) Morphosyntactic operation: construct the 3.sg.pres.act.impv. for 'boil'
($\sqrt{yas}$)

					MAX-IO
/ya-yas-/					
a.	𐬨𐬀	ya-yas-			
b.		yeṣ-		*!	
c.		i-yas-		*!	

/zero grade allomorph to 'boil'/					MAX-IO
	σ_1	σ_2	DEP-V	MAX-IO	
a.	𐬨𐬀	ya-		*	
b.		yas-	*!		
c.		iṣ-	*!	*	

/full grade allomorph to 'boil'/					MAX-IO
	σ_1	σ_2	DEP-V	MAX-IO	
a.	𐬨𐬀	yas-			
b.		iṣ-		*!	
c.		ya-		*!	

Output: *ya-yas-*

Suffix: *-tu*

→ *yayastu*

The interaction of constraints in the Daughter A cophonology demonstrates the outcome for roots with zero grades in τ (23). As in §4.2, I assume that [i] is the unmarked anaptyctic vowel. The validity of this assumption, however, gains further credence in that it helps to account for the formation of reduplicated presents to roots that exhibit *saṃprasāraṇa* in their zero grade, i.e., $\sqrt{vac}$, which has a PPP *uktá-*. If one introduces a further constraint, *ONSET*, which penalizes the formation of syllables without an onset consonant, ranked above *DEP-V*, into the constraint set of the Daughter A cophonology, then the desired reduplicated present form *vívakti* will result (24).¹³

of these conditions hold good, then perhaps a zero-grade allomorph **dadh-* or **dhā-* to $\sqrt{dhā}$ existed at some level of representation. One should also take note of the fact that *dāta-* is the PPP to Avestan $\sqrt{dā-}$ (with conflation of Proto-Indo-Iranian **√dā* 'give' and **√dhā* 'set').

- 13 *vivaṣṭi* to $\sqrt{vaś}$ is likewise an expected outcome under the same account, if the weak stem of the Class II present, e.g., 1.pl.pres.act.ind. *uśmāsi*, is taken as evidence of the proper zero grade allomorph to $\sqrt{vaś}$.

- (23) Morphosyntactic operation: construct the 3.sg.pres.act.ind. for 'bear' ($\sqrt{bhr}$)

/bi-bhar-/					MAX-IO
a.	$\sqrt{bhr}$	bi-	bhar-		
b.		by-	ar-		*!
c.		i-	bhar-		*!

/zero grade allomorph to 'bear' (= bhr)/					MAX-IO
		$*\sigma_\mu > 1$	$*R$	DEP-V	MAX-IO
a.	$\sqrt{bhr}$	bi-		*	*
b.		bhr -		*!	
c.		$bhir$ -	*!	*	

/full grade allomorph to 'bear'/'					MAX-IO
a.	$\sqrt{bhr}$	bhar-			
b.		bhr -			*!
c.		bha-			*!

Output: *bi-bhar-*

Suffix: *-ti*

→ *bibharti*

- (24) Morphosyntactic operation: construct the 3.sg.pres.act.ind. for 'speak' ($\sqrt{vac}$)

/zero grade allomorph to 'speak' (= uc)/					MAX-IO
	ONSET	$*\sigma_\mu > 1$	$*R$	DEP-V	
a.  vi-				*	*
b. uc-	*!	*			
c. u-	*!				*

/full grade allomorph to 'speak'/'					MAX-IO
	ONSET	$*\sigma_\mu > 1$	$*R$	DEP-V	
a.  vac-					
b. va-					*!
c. ac-					*!

Output: *vi-vac-*

Suffix: *-ti*

→ *vivakti*

The presence of a constraint ONSET in the morphologically determined Daughter A cophonology makes for an interesting contrast with Vedic perfects, which also show reduplication: the 3.sg.perf.act.ind. of $\sqrt{vac}$ is *uvāca*, which has the reduplicant the reduplicated present would have produced without ONSET. This contrast, then, perhaps provides evidence of intralinguistic morphologically conditioned phonology.

4.4. Conclusions

Section 4.2 demonstrated that a model that defines reduplication in terms of phonological correspondence between the constituent morphemes, such as BRCT, cannot account for significant parts of the Vedic data. In contrast, MDT allows one to describe the formation of the Vedic reduplicated present as an essentially morphological operation, subject to relatively minor alteration at the phonological level. This outcome bolsters the thesis of Inkelas and Zoll (2005) that reduplication is fundamentally morphological and semantic in nature, rather than phonological.

Although the approach described in §4.3 captures the data of athematic reduplicated presents, certain thematic reduplicated presents, on the other hand, flatly contradict this model, e.g., 3.sg.pres.mid.ind. *jighnate* to $\sqrt{han}$, where the model of §4.3 would predict **jaghmate*. Given that the thematic reduplicated presents are absolutely consistent in the form of the reduplicant regardless of the root phonotactics, one may simply treat them as instantiations of fixed-segment reduplication, as discussed under §3.2.1. Thematic reduplicated presents reflect a morphophonological process distinct from athematic reduplicated presents.

The question of historical origins nevertheless remains: how and why would Vedic have developed the synchronic process that it employs for forming reduplicated presents, in the face of data from other Indo-European languages?

5. Reconstructing the Indo-European reduplicated present

This section will summarize arguments concerning earlier reconstructions of the PIE reduplicated present, then finally make a new attempt, employing MDT to model the reduplicative process.

5.1. One or two reduplicated presents?

In the main, I am sympathetic to the intuitions of Frotscher and Hill (2010:2) that “the assumption of just one stem formation is much more economical” and that “mixed paradigms are better explained ... if *i*- and *e*-reduplication originally belonged to one paradigm,” e.g., *sīṣakti* vs. *sāscati*, *jīgāti* vs. *jāgat-*, or *jāhati* vs. *jihīte*. Models that adopt two types of reduplicated present must appeal to extra-paradigmatic analogy to explain such mixed paradigms, and also have a more difficult burden of proof with respect to semantics.¹⁴

The converse approach, namely, to derive all athematic reduplicated presents in Vedic from a single Indo-European type, goes back at least to Hirt 1900:190–3. More specifically, Hirt suggested that the consistent absence of full grade in the desinence of the 3.pl.pres.act.ind. of reduplicated presents, e.g., *bī-bhr-ati* or *dā-d-ati*, in contrast to the other Vedic athematic present types, which always show full grade of the desinence in the 3.pl.pres.act.ind., e.g., *s-ánti* or *yu-ñ-j-ánti*, points to original accentuation and full grade in the reduplicant; thus Vedic *dā-d-ati* would continue PIE **dé-dh₃-nti*. Watkins 1969:36 makes a similar observation, though Watkins ultimately attributes the full grade of the reduplicant alongside the zero grade of the ending in the 3.pl. to analogy with Vedic intensives of the form 3.pl. *vár-vṛt-ati* (cf. example (7) above).¹⁵ Kortlandt (1987:222) makes essentially the same observation as Hirt, and Kulikov (2005:445–6) follows and expands upon Kortlandt.

Frotscher and Hill (2010) also aim for a single PIE type, but because the Vedic accent is interpreted as absolutely indicative of a PIE full grade, Vedic forms like *bībharti* are derived from PIE **b^hé-b^he/or-ti*. That the reduplicated present took **o*-grade of the root in its strong stem seems improbable, insofar as a stem **b^hé-b^hor-* would then be very similar to the stem of the PIE perfect. Furthermore, the non-operation of Brugmann’s Law in the 3.sg.pres.act. subjunctives *bi-bhar-a-t* and *ba-bhas-a-t*, where an original **o*-grade should have produced **bi-bhār-a-t*, **ba-bhās-a-t*, also speaks against the reconstruction of **o*-grade in the root.¹⁶ For the stem to have been **b^hé-b^her-* would be even stranger, given that

14 The semantic difference that Tichy (2006:113–4) sees between *bībharti* and *dādāhāti* is probably more a feature of lexical semantics than of the reduplication itself.

15 Watkins’ implication seems to be that fixed-segment **i*-reduplication was original to the Indo-European reduplicated present, though a means of accounting for the thoroughgoing *a*-reduplication of *dādāti* and *dādāhāti* is then lacking.

16 Admittedly, however, the data on the inflection of the active subjunctive singular of the reduplicated present is conflicting as to whether full grade or zero grade of the root was original.

PIE did not license the co-occurrence of two *e*-grades in a single word. Hence, the approach of Hirt, Kortlandt, and Kulikov, in which **i*-reduplication is an outcome of zero-grade ablaut, while **e*-reduplication is the outcome of full-grade ablaut, seems most promising.

5.2. Reconstruction and formalization in MDT

I reconstruct the paradigm of the PIE reduplicated present active indicative as follows, using **b^her-* as a sample root:

(25)	<u>Singular</u>	<u>Plural</u>
	<i>*b^hi-b^hér-mi</i>	<i>*b^hi-b^hr-més</i>
	<i>*b^hi-b^hér-si</i>	<i>*b^hi-b^hr-té</i>
	<i>*b^hi-b^hér-ti</i>	<i>*b^hé-b^hr-ṇti</i>
		pres.act.part. <i>*b^hé-b^hr-ṇt-</i>

Thus, one may identify a strong stem in the singular, a weak stem in the first and second plural, and a weakest stem in the third plural and the participle. The root accentuation of the singular is preserved in a few cases, e.g., *bibhárti* and *iyársi*, as well as in all roots with zero grades in *u*.¹⁷ The 1/2.pl., in general, faithfully preserve the original accent and ablaut, whereas the singular has shifted the accent onto the reduplicating syllable (so also Whitney 1889:243), possibly to enhance contrast between the reduplicated present and the perfect. Thus, 3.pl. *sáscati* and pres.part.act. *sáscat*-¹⁸ preserve PIE **sé-sk^w-ṇti* and **sé-sk^w-ṇt-*, respectively,¹⁹ while *jágat*-²⁰ preserves PIE **g^wé-g^wh₂-ṇt-*. Note also the

17 Frotscher and Hill (2010:3) try to dismiss the accentuation of *bibhárti* (RV 4.50.7, AVŚ 19.26.1, 2) and *iyársi* (RV 10.37.4), but both forms are *lectiones difficiliores*, given that the accent throughout the paradigms of those stems is otherwise on the reduplicant. To attribute the accentuation to perfect-derived forms like *cikéši* seems difficult, since such forms are relatively rare (*bibhétī* is not yet present in RV or AV).

18 Jasanoff (2003:129) contends that 3.pl. *sáscati* is a nonce formation, based on the “substantivized participle *saścát-* ‘pursuer’.” That the substantivization of the participle occurred is true enough, but a certainly participial usage, dat.sg. *sáscate*, occurs at RV 2.16.4b. Jasanoff does not account for why *saścát-* itself should have *a*-vocalism in the reduplicant. To simply take *sáscati* at face value seems more straightforward.

19 That the **k^w* of **sé-sk^w-ṇti* and **sé-sk^w-ṇt-* seems to show a palatal outcome (Skt. *c*) in a non-palatalizing environment is unproblematic, because Indo-European root-final (labio-)velars appear as palatals in Skt. when preceding a vowel, regardless of the provenance of that vowel. Put another way, the underlying form of the IE root **sek^w-* has become /sac/ in Skt., so 3.sg. *síṣakti* is the result of internal sandhi from /sísacti/. Compare the RVic paradigm of the Class VII present to √*prc* ‘mix’: 2.sg.act.ind. *prnáksi*, 3.pl.act.ind. *prñcānti*, 3.sg.act.impv. *prñaktu*. An informal count of roots from Whitney 1885 finds seven roots in final *k* versus twenty-two

For the PIE construction, the inputs are taken to be the full-grade form of the root. In accordance with the common view that **e*-grade in PIE bore the word accent, the constraint **E*_[-stress] prohibits an unaccented **e* from surfacing; for my purposes here, I assume that the accent is assigned at a level of derivation preceding what is shown in (26) and (27). Thus, the reduplicant takes zero grade, but **R*_o disallows a syllabic resonant, so this potential syllable nucleus is removed.²¹ The anaptyctic [i] appears at the level of the Mother cophonology in order to satisfy both **GEMINATE* and *MAX-IO*, given that *DEP-V* is ranked lower than those two constraints. Limitations of space here do not permit further comment on the nature of anaptyctic vowels in PIE, but the reader may find helpful discussion in Vine 1999 and Byrd 2010:108. The corresponding weakest stem, in the 3.pl., easily falls out from these constraints as well:

- (27) Morphosyntactic operation: construct the 3.pl.pres.act.ind. for 'bear'
(**b^her-*)

		<i>/b^hér-b^hr-/</i>	<i>*GEMINATE</i>	<i>MAX-IO</i>
a.		<i>b^hér-b^hr-</i>		
b.		<i>b^hér-b^hir-</i>		<i>*!</i>
c.		<i>b^hér-</i>	<i>*!</i>	

		<i>/b^hér-/</i>	<i>*E</i> _[-accent]	<i>*R</i> _o	<i>*CODA</i>	<i>MAX-IO</i>
a.		<i>b^hér-</i>			<i>*!</i>	
b.		<i>b^hé-</i>				<i>*</i>
c.		<i>b^h-</i>				<i>**!</i>
d.		<i>b^hr-</i>		<i>*!</i>		<i>*</i>

		<i>/b^her-/</i>	<i>*E</i> _[-accent]	<i>MAX-IO</i>
a.		<i>b^her-</i>	<i>*!</i>	
b.		<i>b^he-</i>	<i>*!</i>	<i>*</i>
c.		<i>b^h-</i>		<i>**!</i>
d.		<i>b^hr-</i>		<i>*</i>

Output: **b^hér-b^hr-*

Suffix: **-nti*

→ **b^hér-b^hrrnti*

21 **R*_o seems well motivated insofar as no IE language attests to a syllabic resonant or a clear vocalized reflex thereof in a reduplicant. For more on special phonological properties limited to reduplicants, see Inkelas and Zoll 2005:ch.3.

5.3. From PIE to Vedic

The PIE athematic reduplicated present derivations seen in (26) and (27) furnish morphologically identical inputs as the daughters of the reduplicative construction, whereas Vedic, as demonstrated in §4.3, employs morphologically distinct inputs. The final question, then, is: what drove this basic change in the underlying morphological constituency of the reduplicated present between PIE and Vedic? As per Kulikov 2005:445–6, the crucial step appears to lie with laryngeal-final roots (Vedic *Cā*-roots): following the vocalization of the laryngeals to Vedic /i/ when such roots were in the zero grade, speakers began to associate the inherited *i*-vocalism in the reduplicant with the zero-grade allomorph of the root. Consequently, the zero-grade vocalism spread throughout the individual paradigms, thus leading to *u*-reduplication in roots with zero grades in *u* (such as *juhóti*) and *a*-reduplication to roots in which a phonological full grade had come to fill the morphological zero grade, as in roots ending in coronals (such as *bábhasti*).²² $\sqrt{dā}$ and $\sqrt{dhā}$, meanwhile, may have developed a paradigm with *a*-reduplication due to early loss and non-vocalization of the root-final laryngeal; contrast 1.pl.pres. act.ind. *śiśīmasi* with *dadmas* or 3.sg.pres.mid.ind. *śiśīte* with *datté*. $\sqrt{dā}$ and $\sqrt{dhā}$ thereby escaped the pattern of other *Cā*-roots, but ultimately fell victim to the larger pattern that the *Cā*-roots promoted (cf. n12 above).

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22 The constraints and derivational process given in figures (26) and (27) would in fact predict that PIE roots with zero-grade *u*- or *i*-vocalism would have such vocalism in the reduplicant. If one insists that PIE athematic presents with zero grade in *u* never exhibited *u*-vocalism in the reduplicant (note Hittite *kukušzi* < **ḡeus-*), then (26) and (27) require minor modification.

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